

11.1. Generation of Alternating Voltages and Currents

Alternating voltage may be generated by rotating a coil in a magnetic field, as shown in Fig. 11.1 (a) or by rotating a magnetic field within a stationary coil, as shown in Fig. 11.1 (b).

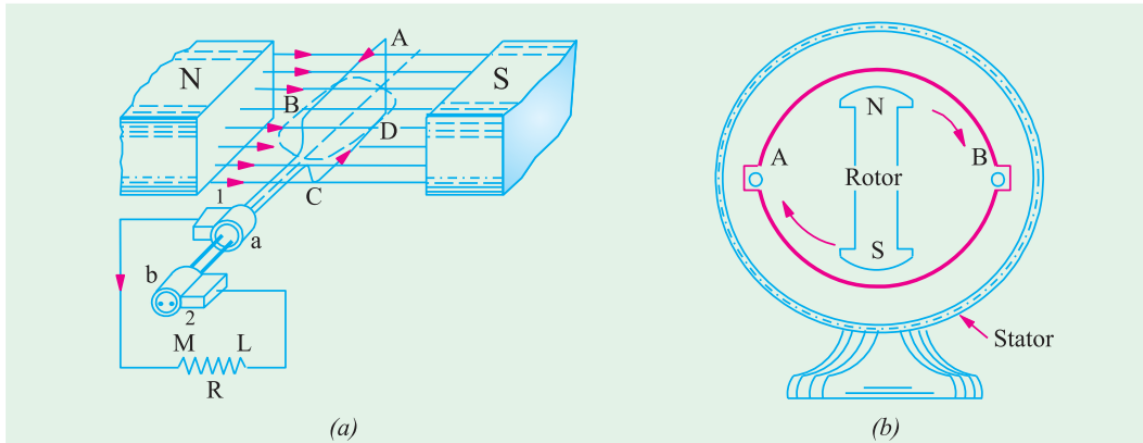


Fig. 11.1

The value of the voltage generated depends, in each case, upon the number of turns in the coil, strength of the field and the speed at which the coil or magnetic field rotates. Alternating voltage may be generated in either of the two ways shown above, but rotating-field method is the one which is mostly used in practice.

11.2. Equations of the Alternating Voltages and Currents

Consider a rectangular coil, having N turns and rotating in a uniform magnetic field, with an angular velocity of ω radian/second, as shown in Fig. 11.2.

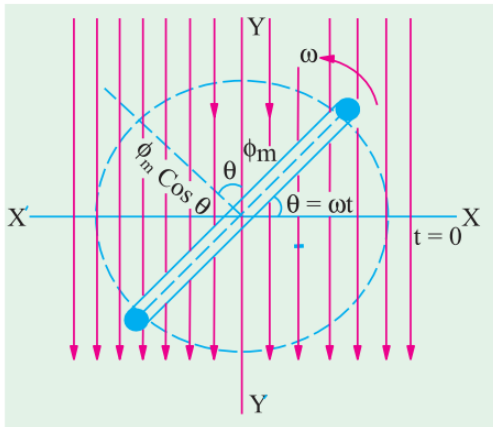


Fig. 11.2

Let time be measured from the X -axis. Maximum flux Φ_m is linked with the coil, when its plane coincides with the X -axis. In time t seconds, this coil rotates through an angle $\theta = \omega t$. In this deflected position, the component of the flux which is perpendicular to the plane of the coil, is $\Phi = \Phi_m \cos \omega t$. Hence, **flux linkages** of the coil at any time are $N \Phi = N \Phi_m \cos \omega t$.

According to Faraday's Laws of Electromagnetic Induction, the e.m.f. induced in the coil is given by the rate of change of flux-linkages of the coil. Hence, the value of the induced e.m.f. at this instant (*i.e.* when $\theta = \omega t$) or the instantaneous value of the induced e.m.f. is

$$e = -\frac{d}{dt}(N \Phi) \text{ volt} = -N \cdot \frac{d}{dt}(\Phi_m \cos \omega t) \text{ volt} = -N \Phi_m \omega (-\sin \omega t) \text{ volt}$$

$$= \omega N \Phi_m \sin \omega t \text{ volt} = \omega N \Phi_m \sin \theta \text{ volt} \quad \dots(i)$$

When the coil has turned through 90° *i.e.* when $\theta = 90^\circ$, then $\sin \theta = 1$, hence e has maximum value, say E_m . Therefore, from Eq. (i) we get

$$E_m = \omega N \Phi_m = \omega N B_m A = 2 \pi f N B_m A \text{ volt} \quad \dots(ii)$$

where

$$B_m = \text{maximum flux density in Wb/m}^2; A = \text{area of the coil in m}^2$$

$$f = \text{frequency of rotation of the coil in rev/second}$$

$$\text{Substituting this value of } E_m \text{ in Eq. (i), we get } e = E_m \sin \theta = E_m \sin \omega t \quad \dots(iii)$$

Similarly, the equation of induced alternating current is $i = I_m \sin \omega t$...**(iv)**
provided the coil circuit has been closed through a resistive load.

Since $\omega = 2\pi f$, where f is the frequency of rotation of the coil, the above equations of the voltage and current can be written as

$$e = E_m \sin 2\pi f t = E_m \sin \left(\frac{2\pi}{T} \right) t \quad \text{and} \quad i = I_m \sin 2\pi f t = I_m \sin \left(\frac{2\pi}{T} \right) t$$

where

T = time-period of the alternating voltage or current = $1/f$

11.10. Different Forms of E.M.F. Equation

The standard form of an alternating voltage, as already given in Art. 11.2, is

$$e = E_m \sin \theta = E_m \sin \omega t = E_m \sin 2\pi f t = E_m \sin \frac{2\pi}{T} t$$

By closely looking at the above equations, we find that

(i) the maximum value or peak value or amplitude of an **alternating voltage is given by the coefficient of the sine of the time angle.**

(ii) **the frequency f is given by the coefficient of time divided by 2π .**

For example, if the equation of an alternating voltage is given by $e = 50 \sin 314t$ then its maximum value of 50 V and its frequency is $f = 314/2\pi = 50$ Hz.

11.13. Root-Mean-Square (R.M.S.) Value

The r.m.s. value of an alternating current **is given by that steady (d.c.) current which when flowing through a given circuit for a given time produces the same heat as produced by the alternating current when flowing through the same circuit for the same time.**

$$I = \sqrt{\left(\frac{i_1^2 + i_2^2 + \dots + i_n^2}{n} \right)}$$

$$V = \sqrt{\left(\frac{v_1^2 + v_2^2 + \dots + v_n^2}{n} \right)}$$

the r.m.s. value of the alternating current is

$$I = \sqrt{\left(\int_0^{2\pi} \frac{i^2 d\theta}{2\pi} \right)} = \sqrt{\left(\frac{I_m^2}{2\pi} \int_0^{2\pi} \sin^2 \theta d\theta \right)} \quad (\text{put } i = I_m \sin \theta)$$

$$\text{Now, } \cos 2\theta = 1 - 2 \sin^2 \theta \quad \therefore \sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\begin{aligned} \therefore I &= \sqrt{\left(\frac{I_m^2}{4\pi} \int_0^{2\pi} (1 - \cos 2\theta) d\theta \right)} = \sqrt{\left(\frac{I_m^2}{4\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{2\pi} \right)} \\ &= \sqrt{\frac{I_m^2}{4} \cdot 2} = \sqrt{\frac{I_m^2}{2}} \quad \therefore I = \frac{I_m}{\sqrt{2}} = 0.707 I_m \end{aligned}$$

Hence, we find that for a symmetrical sinusoidal current

r.m.s. value of current = 0.707 × max. value of current

11.17. Average Value

The average value I_a of an alternating current is expressed *by that steady current which transfers across any circuit the same charge as is transferred by that alternating current during the same time.*

$$I_{av} = \frac{i_1 + i_2 + \dots + i_n}{n}$$

The standard equation of an alternating current is, $i = I_m \sin \theta$

$$\begin{aligned} I_{av} &= \int_0^\pi \frac{id\theta}{(\pi - 0)} = \frac{I_m}{\pi} \int_0^\pi \sin \theta d\theta \\ &= \frac{I_m}{\pi} \left| -\cos \theta \right|_0^\pi = \frac{I_m}{\pi} \left| +1 - (-1) \right| = \frac{2I_m}{\pi} = \frac{I_m}{\pi/2} = \frac{\text{twice the maximum current}}{\pi} \end{aligned}$$

$$I_{av} = I_m / \frac{1}{2} \pi = 0.637 I_m \quad \therefore \text{average value of current} = 0.637 \times \text{maximum value}$$

11.18. Form Factor

It is defined as the ratio, $K_f = \frac{\text{r.m.s. value}}{\text{average value}} = \frac{0.707 I_m}{0.637 I_m} = 1.1$. (for sinusoidal alternating currents only)

In the case of sinusoidal alternating voltage also, $K_f = \frac{0.707 E_m}{0.637 E_m} = 1.11$

11.19. Crest or Peak or Amplitude Factor

It is defined as the ratio $K_a = \frac{\text{maximum value}}{\text{r.m.s. value}} = \frac{I_m}{I_m/\sqrt{2}} = \sqrt{2} = 1.414$ (for sinusoidal a.c. only)

For sinusoidal alternating voltage also, $K_a = \frac{E_m}{E_m/\sqrt{2}} = 1.414$